

Shubhankar Agrawal

Curiosity-driven technologist, passionate about building with data and solving impactful challenges.

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Experience

AIML Engineer, Barclays – Singapore

Dec 2025 – Present

- Building centralized AI platforms for Global Markets enabling workflows covering Sales and Trading use cases.

AI Engineer, GIC – Singapore

May 2025 – Dec 2025

- Engineered data pipelines and agentic tools on AWS to prototype a Research Assistant for Investment Committees.
- Upgraded low-code AI platforms, enabling Private Equity teams to automate and streamline post-trade workflows.

Data Scientist - Machine Learning Engineer, OCBC Bank – Singapore

Jan 2023 – Apr 2025

- Developed recommendation systems for real-time personalization with distributed frameworks generating a 3x uplift in conversions across marketing campaigns and a \$2 million revenue increase.
- Modernized investment advisory creating a Large Language Model (LLM) agent pipeline using LangChain with a RAG-powered text summarization tool, multithreaded to analyse 2000+ research documents across financial assets.
- Led a team to engineer a standardized Python Machine Learning utilities framework for the AI Lab, integrating big data, feature engineering and modelling libraries with scalable I/O, accelerating development timelines by 50%.
- Built solutions to serve models using Flask and modules for model monitoring, back-testing and A/B testing.
- Implemented Forex models: identifying optimal trades for business banking and automated hedging for profitability.

Software Engineer, JP Morgan Chase – Hong Kong

Aug 2020 – Dec 2021

- Delivered new features and enhancements as a server-side developer for a suite of advisor CRM applications, utilizing asynchronous (Reactive) Java, SpringBoot, Kafka and the Elastic stack in biweekly agile sprints.
- Streamlined technical support creating an innovative NLP solution with an analytics dashboard, to auto-reconcile data discrepancies, saving 10 hours of manual intervention weekly increasing operational efficiency.
- Won prizes at internal hackathons and, as an Ignite APAC ML Community Lead organized technical sharing sessions.
- Built a Capture the Flag for 100+ technologists using Kubernetes on Amazon Web Services (AWS) cloud platforms.

Projects

Smart Segment	Optimization based customer segmentation to maximize revenue	pypi.org/project/smart-segment/
HypoRS	Hypothesis testing library with Polars for statistical tests	crates.io/crates/hypors/
Options Pricing	Rust toolkit for Options Pricing and Greeks calculations	github.com/astronights/options-rusty
A4 Trading	Automated trading system to manage a BTC portfolio on tick data	github.com/astronights/A4-Trading
1 Pixel Attacks	Adversarial image generation for computer vision neural networks	arxiv.org/abs/2205.02116
InstaMark LLM	Media captions with Generative AI to boost engagement & marketing	github.com/astronights/insta-llm
Cryptle	Cryptic Crossword built with MongoDB, Express, React, Node, Typescript	daily-cryptic-iief.vercel.app

Education

National University of Singapore – Master of Computing [Artificial Intelligence]

Jan 2022 - Jan 2023

GPA: 4.65 / 5.00 (Awards: Postgraduate AI Scholarship)

University of Hong Kong – Bachelor of Engineering [Computer Science]

Sep 2016 - Jun 2020

GPA: 3.72 / 4.30 (Awards: First Class Honours, Dean's Honour List)

Minors: Finance, French

Skills

AIML: PyTorch, scikit-learn, LLM APIs, LangChain, Hugging Face, Vector DB, Time Series, NLP, Deep learning, A/B testing

Data: Spark, Kafka, Hadoop, Hive, PostgreSQL, Elasticsearch, NumPy, Pandas, Power BI

Programming Languages: Python, R, SQL, JavaScript, Java

| **Tools:** Git, Linux, AWS, Docker, Jupyter, Excel

Languages: English, Hindi, French, Tamil

| **Interests:** Quizzing, Puzzles, Ukulele, Art History